DU Mathsoc Problem Solving: Set 7

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∼Integration Advanced Paper∼

For any assistance or inquiries about the problem sets, please email me at: quizmaster.maths@csc.tcd.ie.

Problem 1. Evaluate $\displaystyle\int_0^1 \frac{\log(1-x)}{x}dx$.

Problem 2. Compute $\displaystyle\int_0^\infty \frac{x}{e^x-1}dx$.

Problem 3. Show that $\displaystyle\int_0^1 x^{-x}dx = \sum_{n\geq 1} n^{-n}$.

Problem 4. Let $f : [0,1] \to \mathbb{R}$ be continuous. Show that $\displaystyle\int_0^1\int_0^1 |f(x)+f(y)|dxdy \geq \int_0^1 |f(x)|dx$.

Problem 5. Let $f : [0,+\infty) \to \mathbb{R}$ be a strictly decreasing continuous function such that $\lim_{x\to\infty} f(x) = 0$. Prove that $\displaystyle\int_0^\infty \frac{f(x)-f(x+1)}{f(x)}dx$ diverges.